1. Title:

Metro Station Connectivity Using an Adjacency Matrix in C++.

2. Problem Statement:

To develop a C++ program that represents metro stations as vertices and direct track connections as undirected edges. The program uses an adjacency matrix to store the graph, accepts station connections from the user, displays the adjacency matrix and identifies all stations directly connected to a selected station.

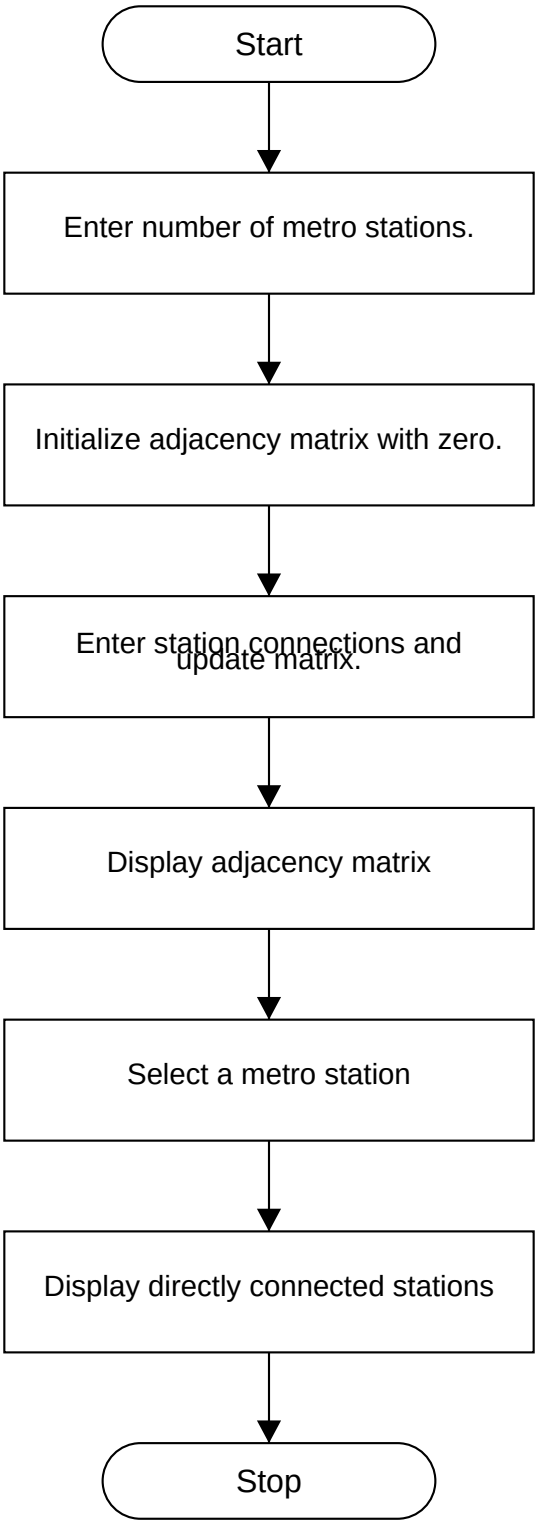
3. Objectives:

- To represent metro stations using vertices in a graph.
- To represent direct metro track connections using undirected edges.
- To implement a graph using an adjacency matrix.
- To display stations directly connected to a selected station.
- To display the adjacency matrix of metro stations.

4. Algorithm:

1. Start the program.
2. Accept the number of metro stations.
3. Assign a name or number to each station.
4. Initialize the adjacency matrix with zero.
5. Accept the number of direct connections between stations.
6. For each connection, set matrix[i][j] = 1 and matrix[j][i] = 1.
7. Display the adjacency matrix.
8. Accept the station number selected by the user.
9. Check the corresponding row of the matrix.
10. Display all stations whose matrix value is 1.
11. Stop the program.

5. Flowchart:


Start
Enter number of metro stations.
Initialize adjacency matrix with zero.
Enter station connections and update matrix.
Display adjacency matrix
Select a metro station
Display directly connected stations
Stop

6. Code:

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
  int n, edges;

  cout << "Enter number of metro stations: ";
  cin >> n;

  vector<vector<int>> adj(n, vector<int>(n, 0));

  cout << "Enter number of direct connections: ";
  cin >> edges;

  cout << "Enter connections using station numbers (1 to "
     << n << "):\n";

  for (int i = 0; i < edges; i++) {
    int u, v;
    cin >> u >> v;

    if (u >= 1 && u <= n && v >= 1 && v <= n) {
      adj[u - 1][v - 1] = 1;
      adj[v - 1][u - 1] = 1;
    } else {
      cout << "Invalid station number!\n";
      i--;
    }
  }
```

```
cout << "\nAdjacency Matrix:\n";

for (int i = 0; i < n; i++) {
   for (int j = 0; j < n; j++) {
      cout << adj[i][j] << " ";
   }
   cout << endl;
}

int station;
cout << "\nEnter station number to find connections: ";
cin >> station;

if (station < 1 || station > n) {
   cout << "Invalid station number!\n";
} else {
   cout << "Stations directly connected to station "
        << station << ": ";

   bool found = false;

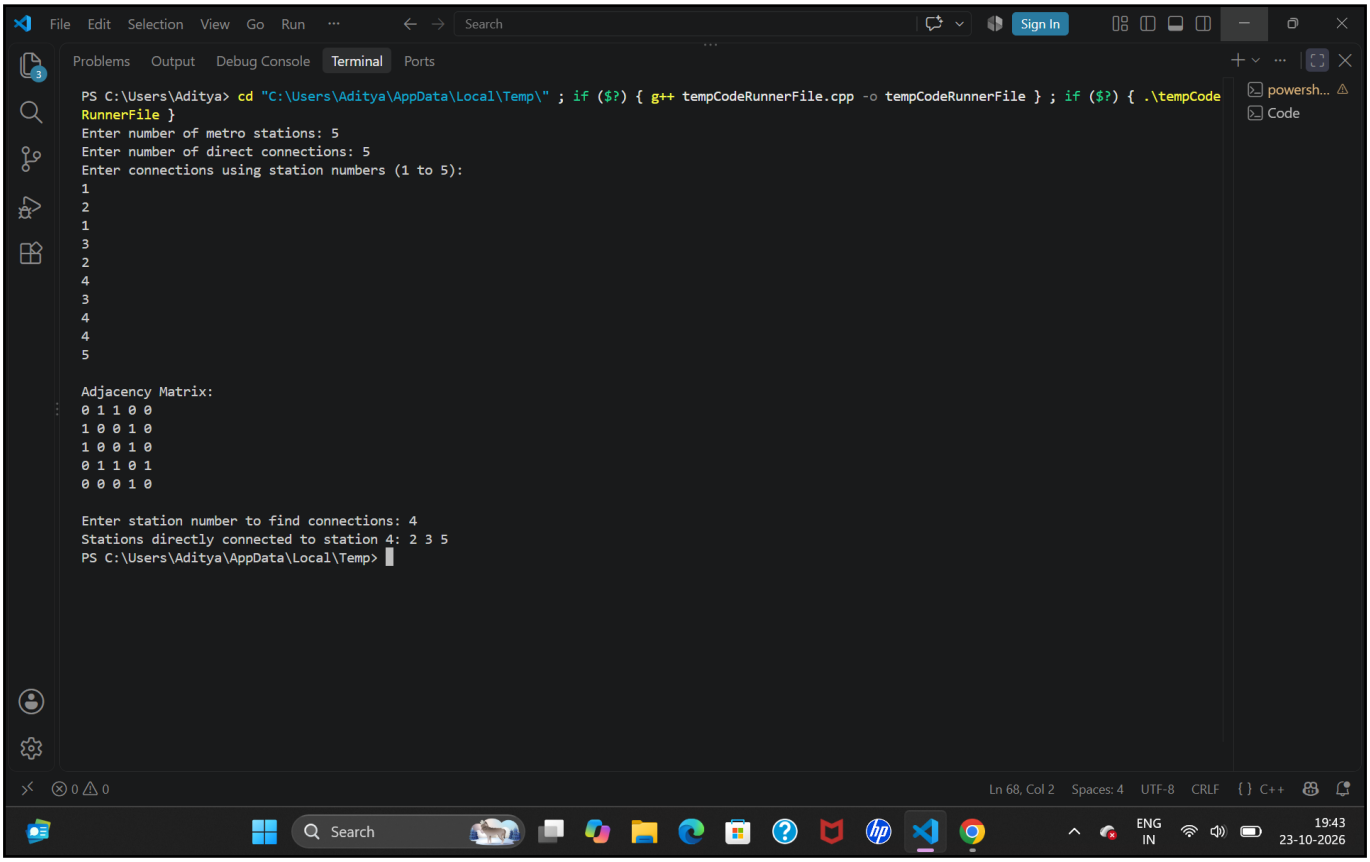
   for (int j = 0; j < n; j++) {
      if (adj[station - 1][j] == 1) {
         cout << j + 1 << " ";
         found = true;
      }
   }

   if (!found) {
      cout << "No direct connections";
   }
```

```
        cout << endl;
    }


    return 0;
}
```

7. Output:


File Edit Selection View Go Run
Search
Sign In
Problems Output Debug Console Terminal Ports
PS C:\Users\Aditya> cd "C:\Users\Aditya\AppData\Local\Temp\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCode
RunnerFile }
Enter number of metro stations: 5
Enter number of direct connections: 5
Enter connections using station numbers (1 to 5):
1
2
1
3
2
4
3
4
4
5
Adjacency Matrix:
0 1 1 0 0
1 0 0 1 0
1 0 0 1 0
0 1 1 0 1
0 0 0 1 0
Enter station number to find connections: 4
Stations directly connected to station 4: 2 3 5
PS C:\Users\Aditya\AppData\Local\Temp>
powersh...
Code
Ln 68, Col 2 Spaces: 4 UTF-8 CRLF { } C++
Search
ENG
IN
19:43
23-10-2026